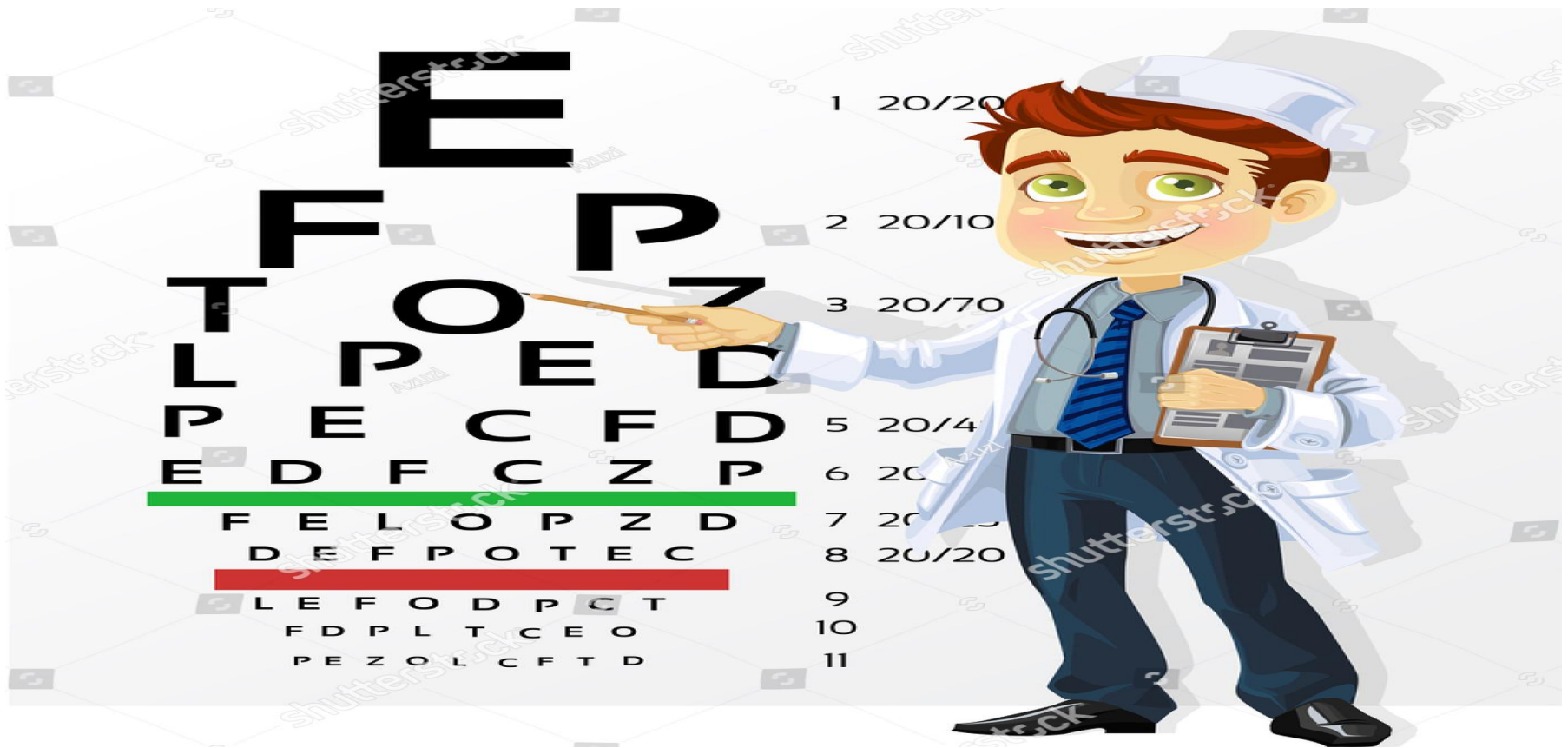


“VISUAL ACUITY AND ERRORS OF REFRACTION”



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“LEARNING OBJECTIVES”

- At the end of lecture student must be able to know,
- What is visual acuity,
- Snellen chart,
- Physiology of Normal vision,
- Normal visual acuity,
- Other measures of visual acuity,
- Refractive errors,
- Causes of refractive errors,
- Management

Visual Acuity

Visual acuity refers to the sharpness of one's eyesight

OR

Visual acuity (VA) is a measure of the ability of the eye to distinguish shapes and the details of objects at a given distance.

It is important to assess VA in a consistent way in order to detect any changes in vision.

Visual Acuity is the clarity or sharpness of vision.

One eye is tested at a time.

Visual Acuity is the capacity to detect, resolve or recognize the details of objects

Visual acuity (VA)

Acuteness or clearness of vision.

It dependent on

- the sharpness of the retinal focus within the eye.
- the sensitivity of the interpretative faculty of the brain.



“VISUAL ACUITY”

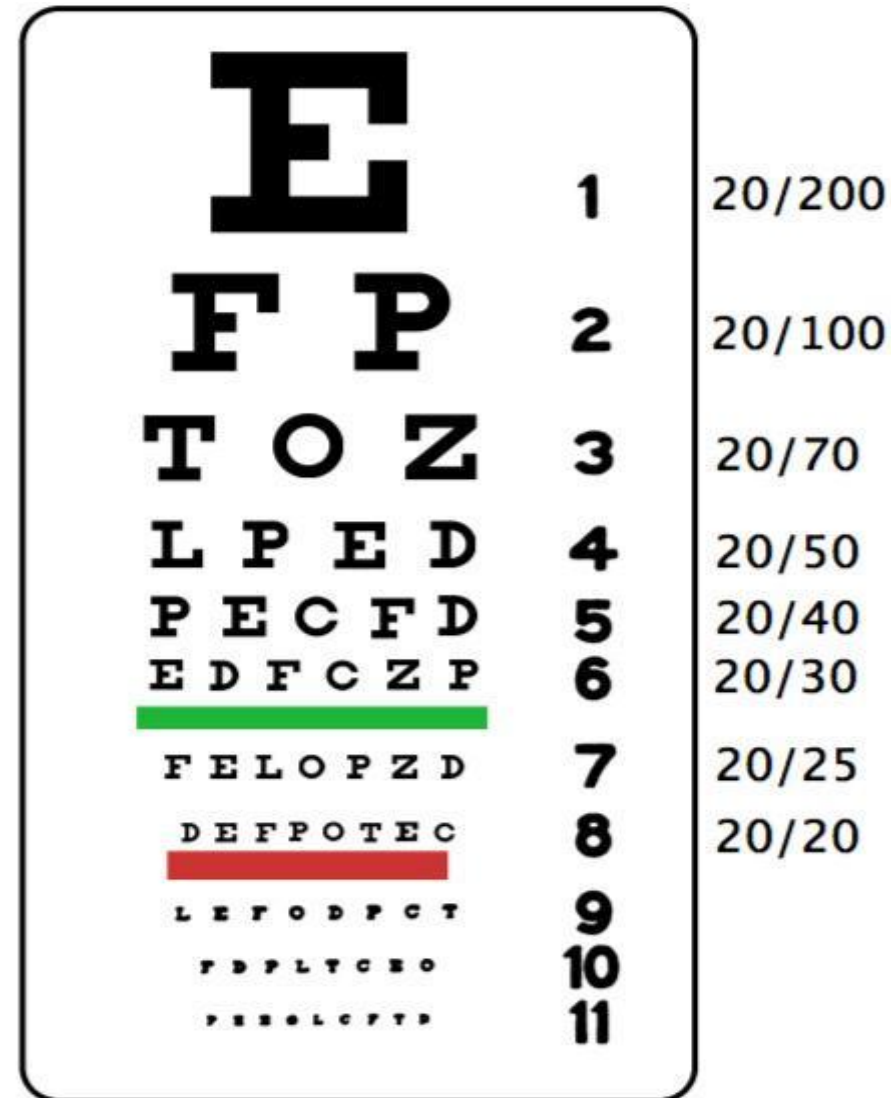
- In 1862 Snellen and Giraud-Toulon introduced a system for measuring VA
- To test visual acuity a **black symbols** on a **white background** are used (for maximum contrast)
- A sufficient distance allowed to approximate infinity in the way the lens attempts to focus.
- **Six meters or twenty feet** is essentially infinity from an optical perspective

“VISUAL ACUITY”

- An eye examination with lenses of varying powers are used to precisely correct for refractive errors.
- Letters are normally used (as in the classic Snellen chart) as most people will recognize them but other symbols (such as a letter E, facing in different directions) can be used instead.

The traditional way to measure vision and visual acuity is to use an Optometrist's test chart, which contains letters of different sizes.

On these charts, the letter size is specified not by its height, but by the distance



“VISUAL ACUITY”

6/6 Vision - the ability to read the 6 meter letter, at a distance of 6 meters (or 20/20)

For the lowest line that is read by an eye with no refractive error (or the errors corrected) is usually 20 feet or 6 meters.

If you can read the second line from 20 feet away but not the, your visual acuity is 20/100. If you cannot read the first line, your visual acuity is at least 20/200 and potentially worse.

Acuity is reported as a fraction.

Numerator – distance from the subject to the chart.

Denominator – size of the symbol the person can see at 6 meters or 20 feet.

Opto types for measuring visual acuity

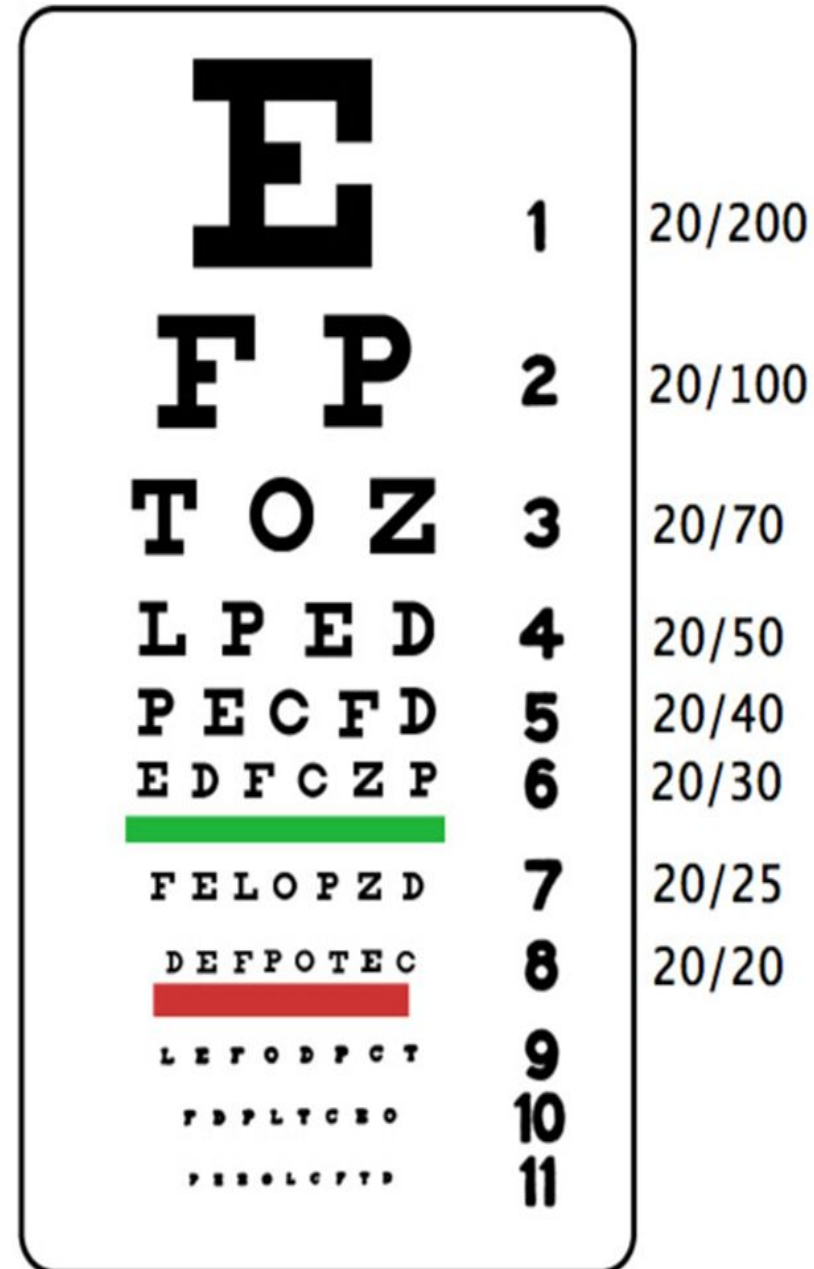
In a clinical practice it is common to use Opto type cards

An opto type card is formed by rows of opto types of progressively smaller sizes.

They are designed to be presented at a fixed distance and on one side they usually show the visual acuity value belonging to each row.

Visual Acuity

- 20/20
 - ability to see letters of a given size at 20 feet
- 20/40
 - what a normal person can see at 40 feet, this person must be at 20 feet to see.
- 20/200
 - what a normal person can see at 200 feet, this person must be at 20 feet to see.

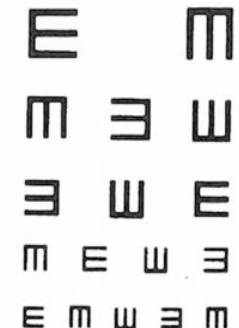
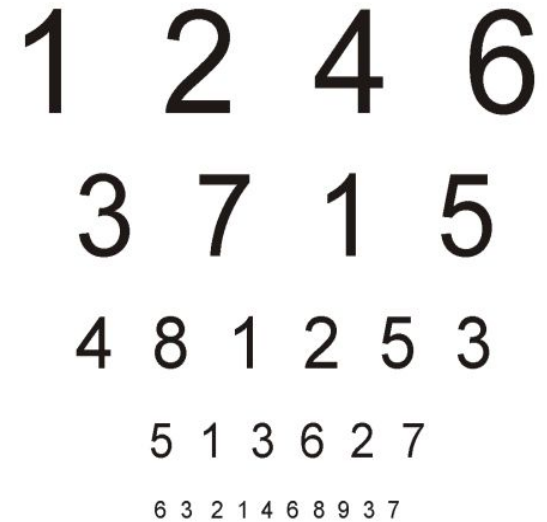
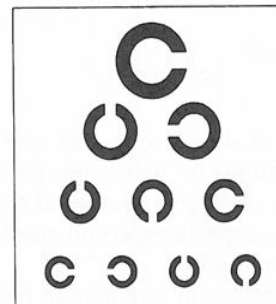


Types of (Optotypes): Visual Acuity

Types of test

(Optotypes):

- Letters
- Numbers
- Snellen "E"
- Landolt Rings



Optotypes for measuring visual acuity

✓ Optotype cards

Letter Cards

Snellen

E	20/200
L T	20/100
F P H	20/70
O L C F	20/50
D H J B S	20/40
E P T Z O	20/30
C F D H J	20/25
L T I P H	20/20

H	60
P N	36
D Z U	24
F R V E	18
Z H N U D	12
V P D E F R	9
P R E U H D N Z	6
U V D H E N F P	5
R U Z P N H D F	4
E D N Z F H H P U	3

Optotypes for measuring visual acuity

✓ Optotype Cards

Letter Cards

Bailey-Lovie



Optotypes for measuring visual acuity

✓ Optotype Cards

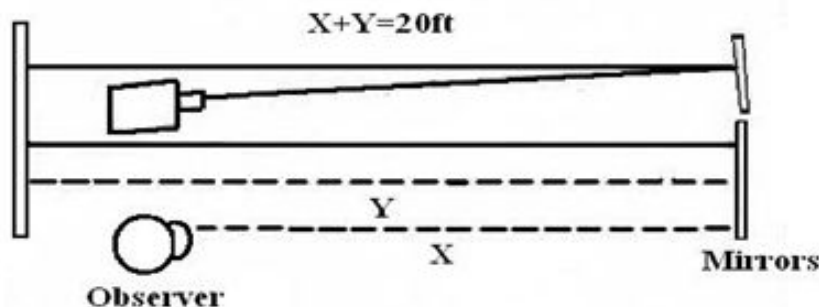
Letter Cards

Optotype Projectors



self illuminated optotypes,

- distances between 10ft and 20ft
- Optically “fold” the testing distance using mirrors to create appropriate testing distance (20ft is the standard)



Optotypes for measuring visual acuity

✓ Optotype Cards

Number Cards

Optotype Projectors

1 2 4 6
3 7 1 5
4 8 1 2 5 3
5 1 3 6 2 7
6 3 2 1 4 6 8 9 3 7

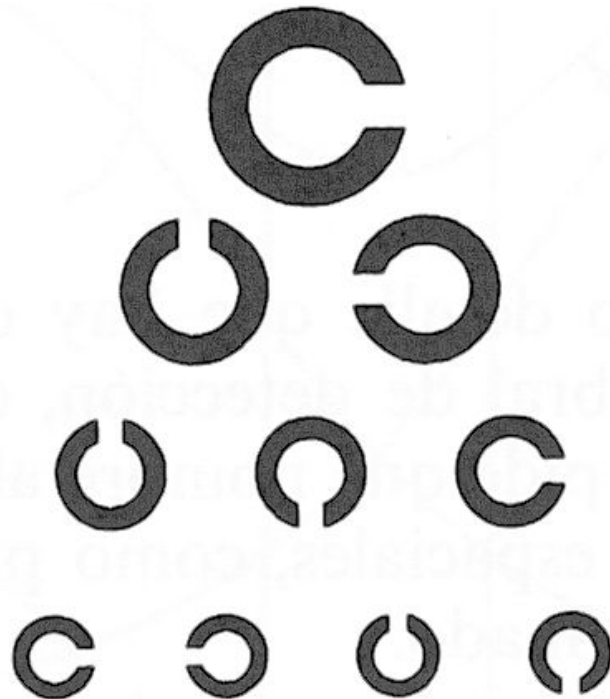


Optotypes for measuring visual acuity

✓ Optotype Cards

One Character Optotype Cards

Landolt Rings



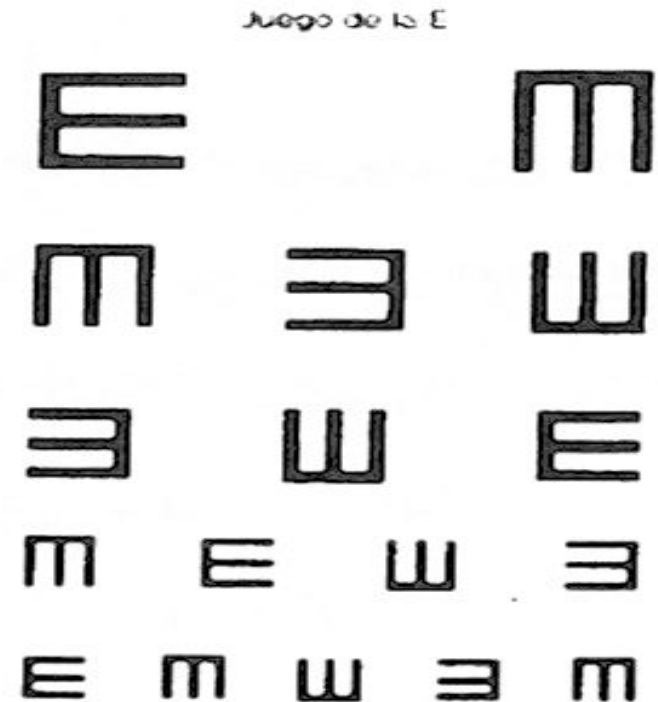
Optotypes for measuring visual acuity

✓ Optotype Cards

One Character Optotype Cards

Snellen "E"

Tumbling E Chart



Optotypes for measuring visual acuity

✓ Optotype Cards

Other Cards

Infant Visual Acuity



Visual Acuity Testing

- Use Snellen chart
 - Positioned 6 meters or 20 feet away
- Each Eye Alone, Then Together
- With Corrective Lenses (If indicated)

Testing Distance Visual Acuity

General Procedures:

- Make sure the room is well lit.
- Choose appropriate chart.
- Chart at eye level.
- Mark off appropriate distance for chart.
- Patient position their heels on the line.
- Ask if the person wears glasses, if yes, testing should be done with the person wearing the glasses. (Be sure to mark the form)

Testing Distance Visual Acuity (cont.)

- Watch carefully to observe if the person is peeking, tilting their head, or squinting.
- Instruct the person to keep both eyes open and read the selected letter or line of letters with the uncovered eye. Do not use a marking device as a pointer.
- When screening, start at the 1st line and move down to the last line. If the person is unable to read lower lines, move upward.
- Record results including the line number for the last line read correctly with each eye. All failures should be rechecked.

Testing Near Visual Acuity

- Ask if the person wears glasses. If so, screen with the glasses on.
- Hold the card at the appropriate distance. (14-16 inches)
- Do not allow the person to lean the head or trunk forward.
- Watch carefully to be sure the person is not peeking, tilting the head, or squinting.
- Occlude the left eye with an occlude (a card, palm of the hand, or cup) and test the right eye.

Testing Near Visual Acuity (cont.)

- Ask the person to name or read the letter or symbols on each line as directed.
- If the person is unsuccessful, he or she should be rescreened.
- Record results.

Watch Out!



- Watch for peeking

* Resource – MA Preschool Vision Screening Protocol

Factor that affect Visual Acuity

Factors affecting visual acuity, or the sharpness of vision, include

- **Optical factors** like refractive errors (nearsightedness, farsightedness, astigmatism) and media opacities (e.g., cataracts)
- **Neural factors** such as the health of the retina, optic nerve, and brain's visual pathways.
- **Environmental conditions** like lighting, contrast, and glare also play a significant role, as do personal factors like age, attention, fatigue, and pupil size

Refractive Errors

- Myopia (near-sighted)
- Hyperopia (far-sighted)
- Presbyopia:
- Astigmatism:
- Anisometropia

Refractive errors

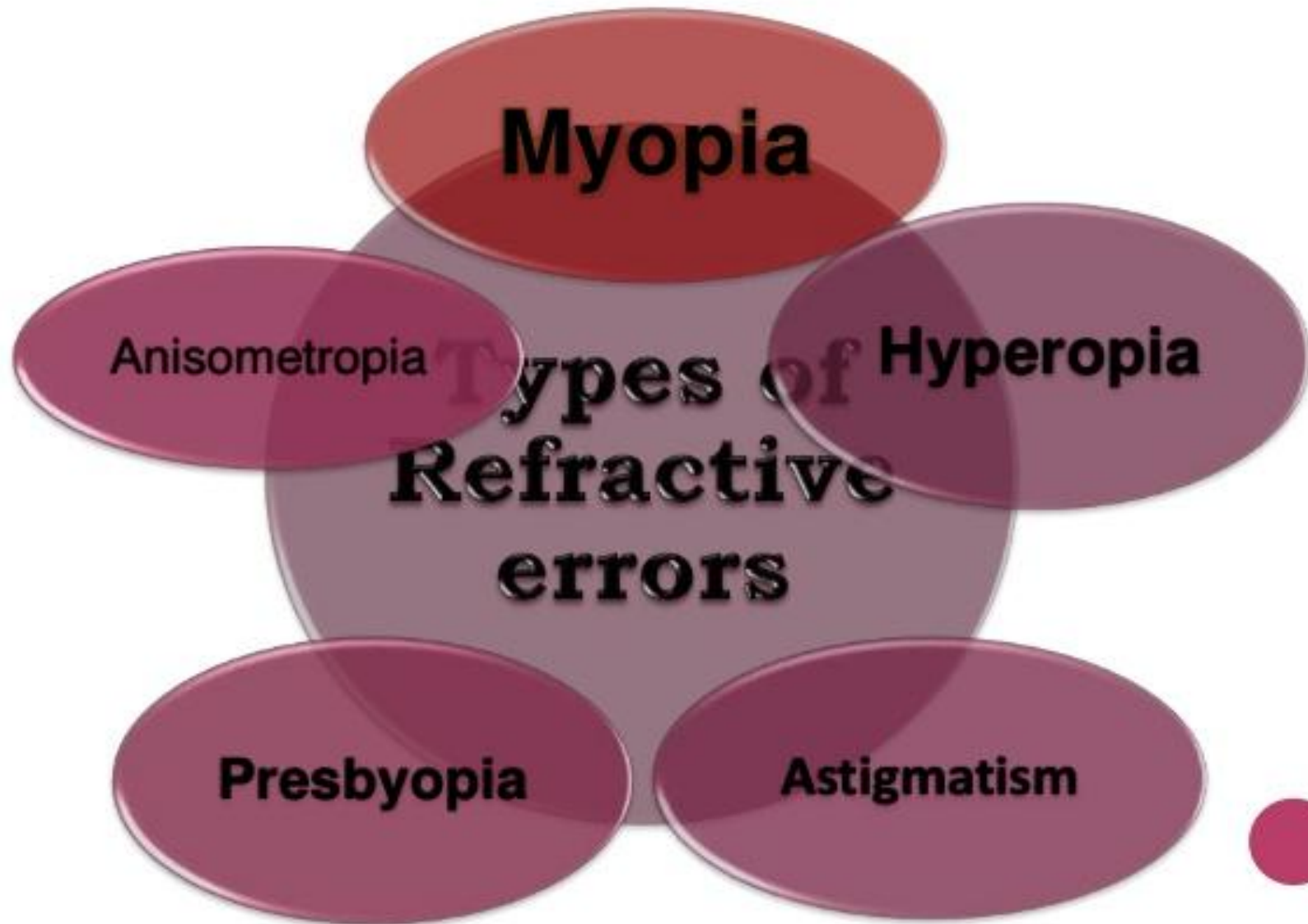
- Failure of parallel light rays to come to point focus on the retina
- A frequent reason for reduced visual acuity.

Emmetropia (normal vision)

An eye that has no refractive error when viewing a distant object is said to have *emmetropia* or be *emmetropic*

Ametropia:

A refractive error is present



Myopia (near-sighted)

- Inability of the lens to accommodate adequately for far vision
- image is focused in front of retina
- occurs when the optics are too powerful for the length of the eyeball (an eyeball that is too long)

Hyperopia (far-sighted)

- Inability of the lens to accommodate adequately for near vision
- image is focused behind retina
- occurs when the optics are too weak for the length of the eyeball (an eyeball that is too short)

Presbyopia:

Presbyopia is when your eyes gradually lose the ability to see things clearly up close. It is a normal part of aging.

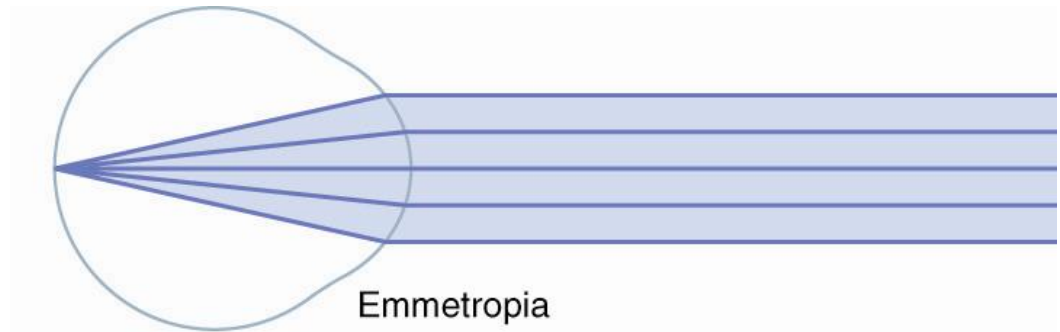
Astigmatism:

Astigmatism (uh- STIG-muh-tiz-um) is a common and generally treatable imperfection in the curvature of the eye that causes blurred distance and near vision.

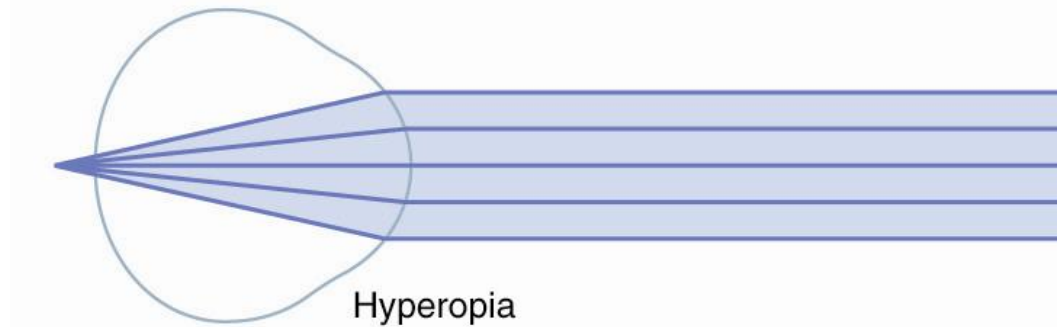
Anisometropia

Difference in the refractive errors of the two eyes.
If sufficiently different □ more blurred image

Normal vision



Far sightedness



Near sightedness

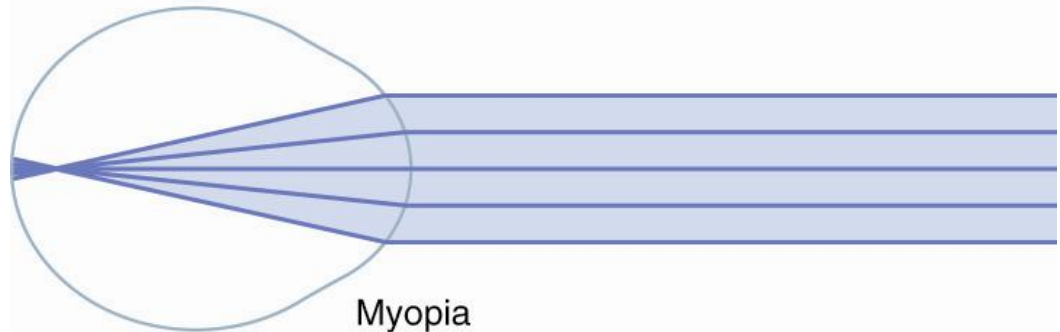


Figure 49-12; Guyton and Hall

Normal Eye



Light rays focus
on the retina

Myopia



Light rays focus
in front of the retina

Astigmatism



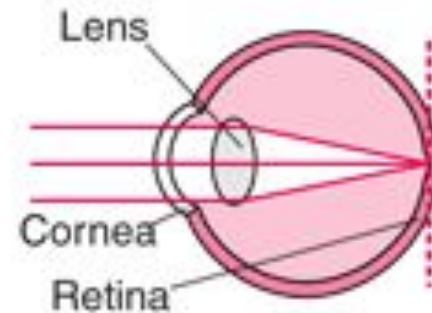
Light rays focus on more than
one point (unequal refraction of
light in different meridians)

Hypermetropia

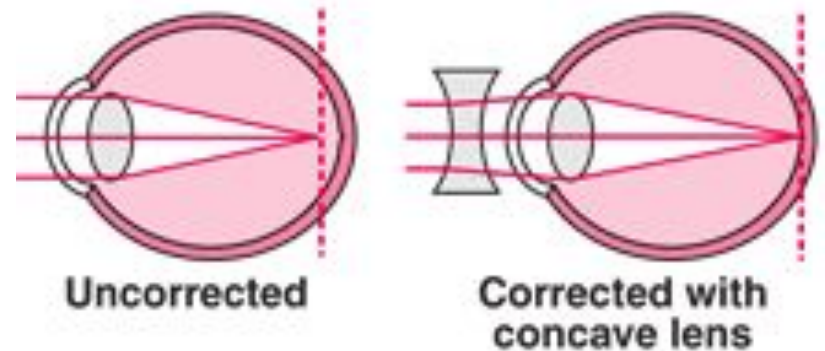


Light rays focus
behind the retina

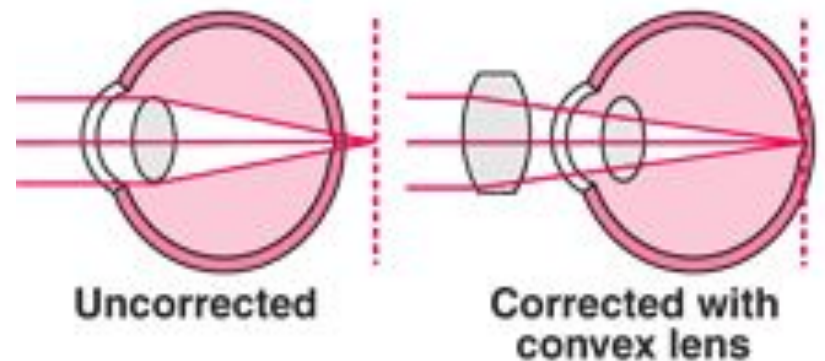
Normal Vision



Nearsighted Vision



Farsighted Vision



Causes

- Refractive errors are thought to occur due to a combination of genetic and environmental factors.
- Trauma or ocular disorders such as keratoconus may induce refractive errors

Management

- How refractive errors are treated or managed depends upon the amount and severity of the condition.
- Those who possess mild amounts of refractive error may elect to leave the condition uncorrected, particular if the patient is asymptomatic.

Management

For those who are symptomatic,

- Glasses,
- Contact lenses,
- Refractive surgery,
- Or a combination of the three are typically used.

Tips to Ease Eye Strain



Sit at arm's length, or 25 inches, from the computer screen.



Follow the "20-20-20" rule.



Adjust your room lighting and increase the contrast on your computer.



Use artificial tears to refresh your eyes when they feel dry.

THE 20-20-20 RULE

to reduce the effects of digital eye strain



Take a break
for **20** seconds...



...and look at something
20 feet away...



...every **20** minutes.

Follow the 20-20-20 rule and give your eyes a break!

healthyssetgo.

Relevant Questions

Q-1-What is visual acuity?

Visual acuity is a measure of the sharpness or clarity of your vision, indicating how well you can distinguish fine details at a specific distance.

Q-2-How is visual acuity is tested?

Visual acuity is tested using a Snellen chart, which has rows of letters or symbols that decrease in size, where the patient stands 20 feet away and reads each line until they can no longer see the symbols clearly

Q-3-What factors influence visual acuity

Factors affecting visual acuity, or the sharpness of vision, include both optical factors like refractive errors (nearsightedness, farsightedness, astigmatism) and media opacities (e.g., cataracts) and neural factors such as the health of the retina, optic nerve, and brain's visual pathways.

Q-4-What is nodal point and what is its significance

The nodal point of the eye refers to a specific point in the optical system of the eye where light rays entering the eye converge. In a simplified model of the eye, the nodal point can be considered as a point through which light rays pass without being refracted, effectively acting like an ideal lens.

Q-5-What is the basis for the standard charts used for testing distant and near vision

Standard eye charts are based on the concept of optotypes (symbols) that subtend a specific visual angle (usually 5 minutes of arc for normal vision) at a set distance (typically 20 feet or 6 meters), with letter sizes decreasing logarithmically down the chart.

Q-6-What do you understand by the visual acuity of 6/6,6/12,6/60

Visual acuity ratios like 6/6, 6/12, and 6/60 on a Snellen chart indicate how well a person can see at a distance of 6 meters (20 feet) compared to a person with standard vision. 6/6 is standard vision, 6/12 means you see at 6m what a standard eye sees at 12m (requiring you to be closer), and 6/60 is considered low vision, as you need to be at 6m to see what a standard eye can see from 60m away.

Q-7-What do you mean by the term short sightedness?How it can be treated

The medical name for short-sightedness is myopia. Eyesight problems, such as myopia, are also known as refractive errors. Short-sightedness leads to blurred vision when looking at things at a distance, whilst close vision is usually normal. Short-sightedness is a very common problem that can be corrected by concave lens , contact lenses, or laser eye surgery

Q-8-What do you mean by the term farsightedness? What causes the eye to be farsighted

Farsightedness (hyperopia) is a vision condition where close objects appear blurry because the eye's lens or cornea doesn't focus light correctly, causing it to be focused behind the retina instead of on it. This happens because the eyeball is too short, the cornea's curve is too shallow, or the lens's focusing power is too weak. Farsightedness is often hereditary and can be present from birth, with children's flexible lenses often compensating for the issue until they age.



HOW TO AVOID EYE STRAIN

WHILE WORKING AT THE COMPUTER



REGULAR
EYE CHECKING



EYES EXERCISING



PROPER
LIGHTING



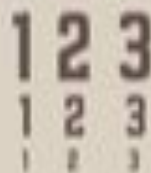
MONITOR
CLEANING



DISTANCE
ADJUSTING



GLARE
REDUCING



FONT SCAING



FREQUENT
BLINKING



GLASSES
WEARING

20-20-20 RULE



20 EVERY
MINUTES



20 BREAK FOR
SECONDS



20 LOOK AT
FEET AWAY